Week 7:

Develop an application that inserts some notifications into Notification area and whenever a notification is inserted, it should show a toast with details of the notification.

Activtity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">


    <Button
        android:id="@+id/btnCreateNotification"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Create Notification"
        tools:layout_editor_absoluteX="170dp"
        tools:layout_editor_absoluteY="322dp" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

Mainactivity.java

```
package com.example.notify;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;

import android.annotation.SuppressLint;
import android.os.Bundle;

/*public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}*/
import android.app.NotificationChannel ;
import android.app.NotificationManager ;
import android.content.ContentResolver ;
import android.content.Context ;
import android.graphics.Color ;
import android.media.AudioAttributes ;
import android.net.Uri ;
//import android.support.v4.app.NotificationCompat ;
//import android.support.v7.app.AppCompatActivity ;
//import android.os.Bundle ;
import android.view.View ;
```

```
import android.widget.Button ;

import java.util.Objects;

public class MainActivity extends AppCompatActivity {
    public static final String NOTIFICATION_CHANNEL_ID = "10001" ;
    private final static String default_notification_channel_id;

    static {
        default_notification_channel_id = "default";
    }

    @Override
    protected void onCreate (Bundle savedInstanceState) {
        super .onCreate(savedInstanceState) ;
        setContentView(R.layout. activity_main ) ;
        @SuppressLint({"MissingInflatedId", "LocalSuppress"}) Button
btnCreateNotification = findViewById(R.id.btnCreateNotification ) ;
        btnCreateNotification.setOnClickListener( new View.OnClickListener()
{
            @SuppressLint("NotificationPermission")
            @Override
            public void onClick (View v) {
                Uri sound = Uri. parse (ContentResolver.
SCHEME_ANDROID_RESOURCE + "://" + getPackageName() +
"/raw/quite_impressed.mp3" ) ;
                NotificationCompat.Builder mBuilder = new
NotificationCompat.Builder(MainActivity. this,
                        default_notification_channel_id )
                        .setSmallIcon(R.drawable. ic_launcher_foreground )
                        .setContentTitle( "Test" )
                        .setSound(sound)
                        .setContentText( "Hello! This is my first push
notification" );
                NotificationManager mNotificationManager =
(NotificationManager) getSystemService(Context. NOTIFICATION_SERVICE ) ;
                if (android.os.Build.VERSION. SDK_INT >=
android.os.Build.VERSION_CODES. O ) {
                    AudioAttributes audioAttributes = new
AudioAttributes.Builder()
                            .setContentType(AudioAttributes.
CONTENT_TYPE_SONIFICATION )
                            .setUsage(AudioAttributes. USAGE_ALARM )
                            .build() ;
                    int importance = NotificationManager. IMPORTANCE_HIGH ;
                    NotificationChannel notificationChannel = new
                            NotificationChannel( NOTIFICATION_CHANNEL_ID ,
"NOTIFICATION_CHANNEL_NAME" , importance) ;
                    notificationChannel.enableLights( true ) ;
                    notificationChannel.setLightColor(Color. RED ) ;
                    notificationChannel.enableVibration( true ) ;
                    notificationChannel.setVibrationPattern( new long []{ 100
, 200 , 300 , 400 , 500 , 400 , 300 , 200 , 400 }) ;
                    notificationChannel.setSound(sound , audioAttributes) ;
                    mBuilder.setChannelId( NOTIFICATION_CHANNEL_ID ) ;
                    assert mNotificationManager != null;
```

```
mNotificationManager.createNotificationChannel(notificationChannel) ;
                }
                Objects.requireNonNull(mNotificationManager).notify(( int )
System. currentTimeMillis (), mBuilder.build()) ;
            }
        }) ;
    }
}
```

output:



